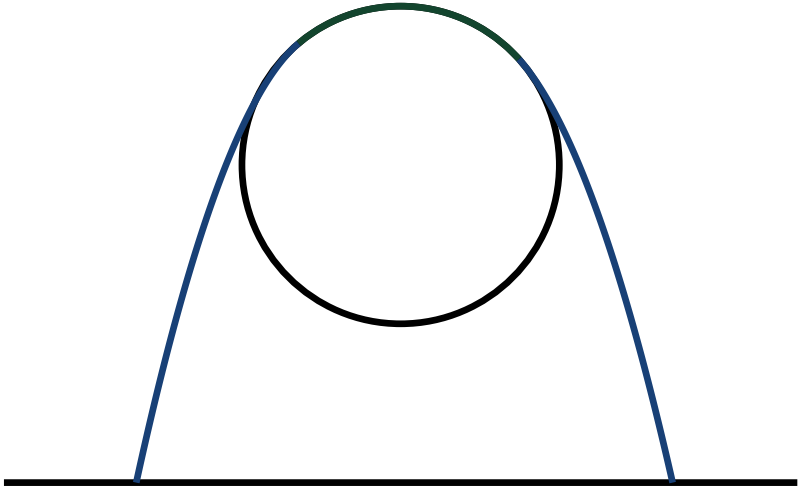
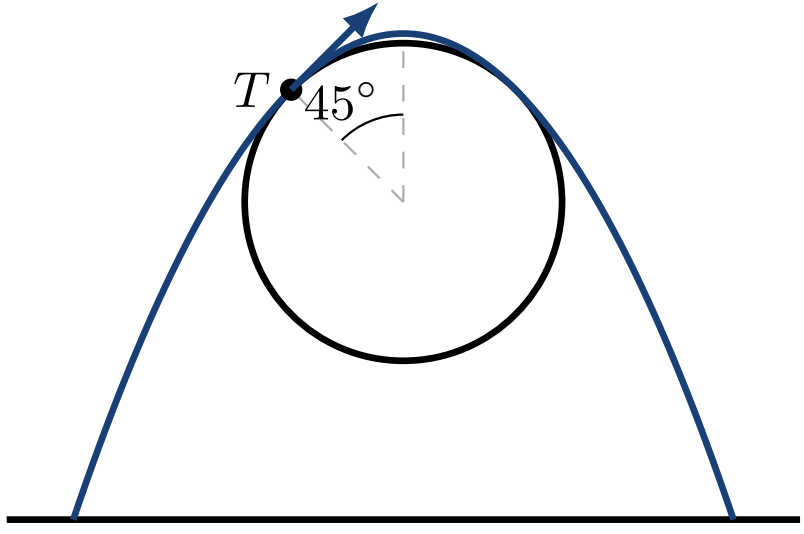
允许接触：势垒 $h + r$



临界速度仅为下确界

$\cos\varphi_{\text{detach}} = 2/3$

禁止接触：极限抛物线



两侧均在 $45°$ 处相切

实际轨迹须留正间隙